

EMOTION DETECTION: Real-Time Facial Emotion Recognition

![[Python]](<https://img.shields.io/badge/Python-3.10.11-blue.svg>)

![[TensorFlow]](<https://img.shields.io/badge/TensorFlow-2.10.0-orange.svg>)

![[Streamlit]](<https://img.shields.io/badge/Streamlit-1.22.0-red.svg>)

![[OpenCV]](<https://img.shields.io/badge/OpenCV-4.13.0-green.svg>)

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Project Overview

EMOTION DETECTION is a lightweight, edge-deployable deep learning system engineered to classify seven universal human emotions (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise) from facial images in real-time.

Built as a Master of Computer Applications (MCA) thesis project, this system leverages a customized MobileNetV2 architecture utilizing a multi-phase transfer learning strategy. It is specifically optimized to run locally on consumer-grade hardware (NVIDIA RTX 3060) without relying on high-latency cloud APIs, achieving sub-100ms inference speeds via native OpenCV hardware hooks and a custom Streamlit UI.

Core Features

Real-Time Edge Inference: Bypasses browser-based latency by utilizing ``cv2.VideoCapture(0)`` for zero-latency optical tracking.

Lightweight Architecture: Utilizes MobileNetV2 with depthwise separable convolutions to drastically reduce parameter count and VRAM footprint.

Dual-Tab UI/UX: Features a custom "Cyberpunk" aesthetic Streamlit dashboard with dedicated modules for live webcam scanning and static image diagnostics.

Advanced Regularization: Trained on the unconstrained FER2013-4X dataset using real-time spatial data augmentation and heavy dropout layers to prevent overfitting.

Peak Performance: Achieves 63.34% validation accuracy, approaching the accepted human-level baseline for the highly ambiguous FER2013 benchmark.

Technology Stack & Environment

To ensure strict scientific reproducibility and hardware stability (specifically native Windows CUDA/cuDNN support without WSL2), this project is locked to the following environment:

Language: Python 3.10.11

Deep Learning: TensorFlow 2.10.0 / Keras

Computer Vision: OpenCV 4.13.0 (Haar Cascades)

Frontend UI: Streamlit 1.22.0

Data Science: NumPy 1.24.3, Pandas, Matplotlib, Seaborn

System Architecture & Directory Structure

EMOTION_DETECTION_PROJECT/

```
|
|
├── dataset/
|   ├── train/      # Contains training images across 7 classes
|   └── test/       # Contains isolated validation images
|
|
├── models/
|   ├── Emotionfinalmodel.h5 # Phase 3 Fine-Tuned Weights
|   ├── Emotionlabels.json  # Categorical Mapping
|   └── haarcascade_frontalface_default.xml
|
|
├── src/
|   ├── eda_analysis.ipynb  # Exploratory Data Analysis & Visualizations
|   └── model_training.py    # Data generators, MobileNetV2 architecture, callbacks
|
|
├── App.py              # Main Streamlit Execution Script
├── requirements.txt     # Environment dependencies
└── README.md           # Project Documentation
```

Installation & Setup

1. Clone the repository and navigate to the project root:

Bash

```
git clone [https://github.com/Vishubhaumick/Emotion-
Detection.git](https://github.com/Vishubhaumick/Emotion-Detection.git)
```

```
cd Emotion-Detection
```

2. Initialize and activate a strict virtual environment:

Bash

```
python -m venv venv
```

On Windows:

```
venv\Scripts\activate
```

On macOS/Linux:

```
source venv/bin/activate
```

3. Install the locked dependencies to prevent version conflicts:

Bash

```
pip install -r requirements.txt
```

4. Verify GPU Allocation (Optional but recommended): Ensure your local CUDA 11.2 and cuDNN 8.1.0 paths are configured correctly for TensorFlow 2.10.0 to recognize your GPU.

Execution

To launch the EMOTION DETECTION dashboard, run the master execution script:

Bash

```
streamlit run App.py
```

Note: Ensure your physical webcam is not currently locked by another application (e.g., Zoom, Teams) before initializing the Real-Time Sensor tab.

Evaluation & Metrics

The multi-phase fine-tuning strategy yielded the following progression on the mathematically isolated validation vault:

- **Phase 1 (Frozen Base):** ~47.99% Accuracy
- **Phase 2 (Mid-Level Unfreeze):** ~59.11% Accuracy
- **Phase 3 (Deep Fine-Tuning): 63.34% Peak Accuracy** (Epoch 23)

Comprehensive diagnostic charts, including Confusion Matrices and Epoch progression graphs, are available in the project report documentation.